

A

 LV.0 → LV.1 · AI ENGINEER

The 8 A's that move you from **Level 0** → **1** AI engineer.

ANN, Agentic AI, AUC-ROC and more. **Plain-English** definitions, one swipe each, plus the agentic IDE everyone's talking about and six people to follow.

ARTIFICIAL NEURAL NETWORK • FOUNDATIONS

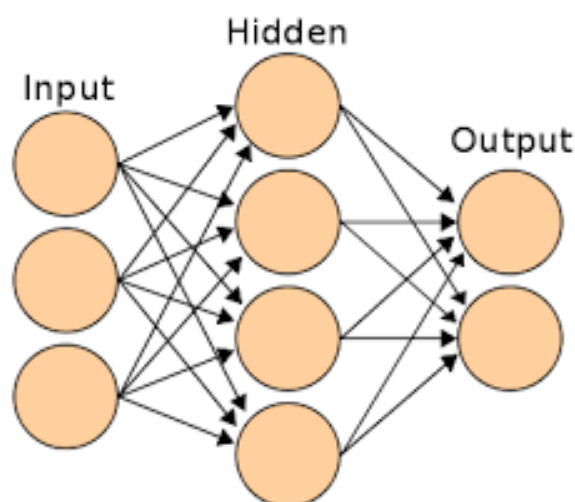
ANN

input • hidden • output

An Artificial Neural Network (ANN) is a computational model inspired by biological neurons. It consists of **interconnected layers of nodes** that learn patterns from data through weighted connections, enabling it to solve complex tasks like classification and prediction.

USED IN

- Deep learning
- Classification
- Prediction



LAYERED NEURONS

ACTIVATION FUNCTION • FOUNDATIONS

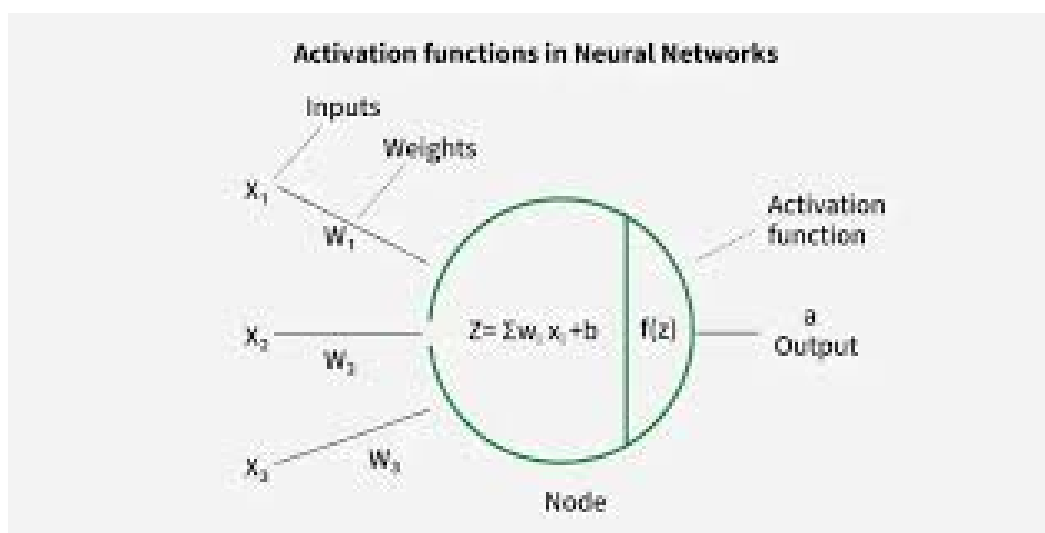
Activation

$z \rightarrow f(z) \rightarrow \text{output}$

An activation function is a mathematical operation in neural networks that **introduces non-linearity**, allowing the network to learn complex patterns. Common ones include ReLU, sigmoid, and tanh. Without them, deep networks would collapse into linear models.

USED IN

- Every deep net
- ReLU / Sigmoid
- Hidden layers



ADDS NON-LINEARITY

ALPHA • LEARNING RATE • TRAINING

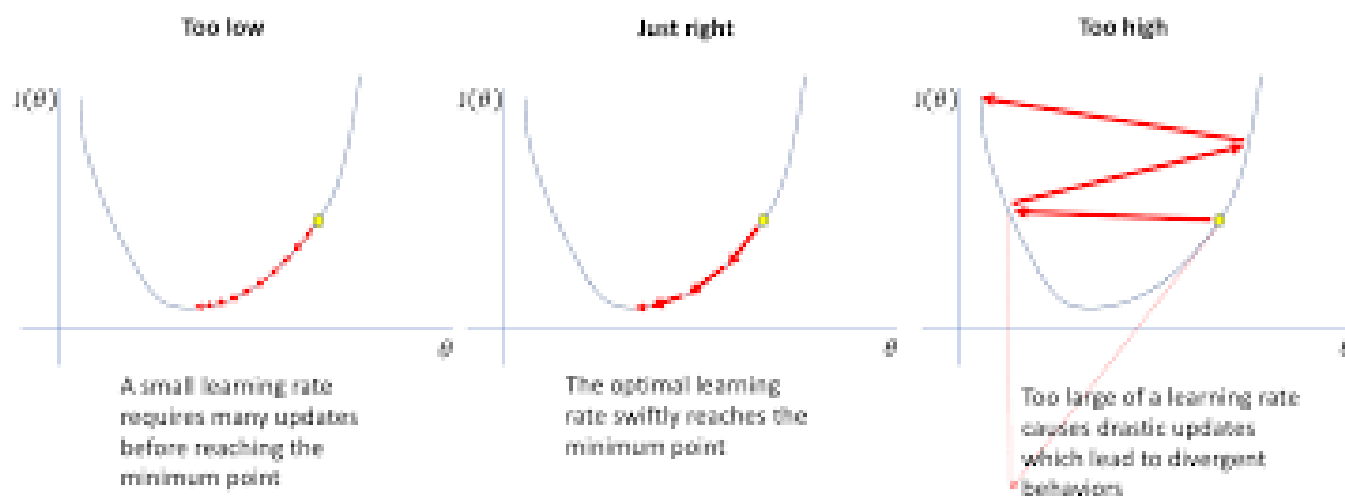
Alpha

step size per update

Alpha (learning rate) is a hyperparameter controlling **how much a neural network's weights update** during training. Higher alpha means faster learning but risks overshooting the optimal solution, while lower alpha offers stability but slower convergence.

USED IN

- Gradient descent
- Optimizers
- Hyperparameter tuning



TOO LOW • JUST RIGHT • TOO HIGH

AUTONOMOUS AGENT • AGENTS

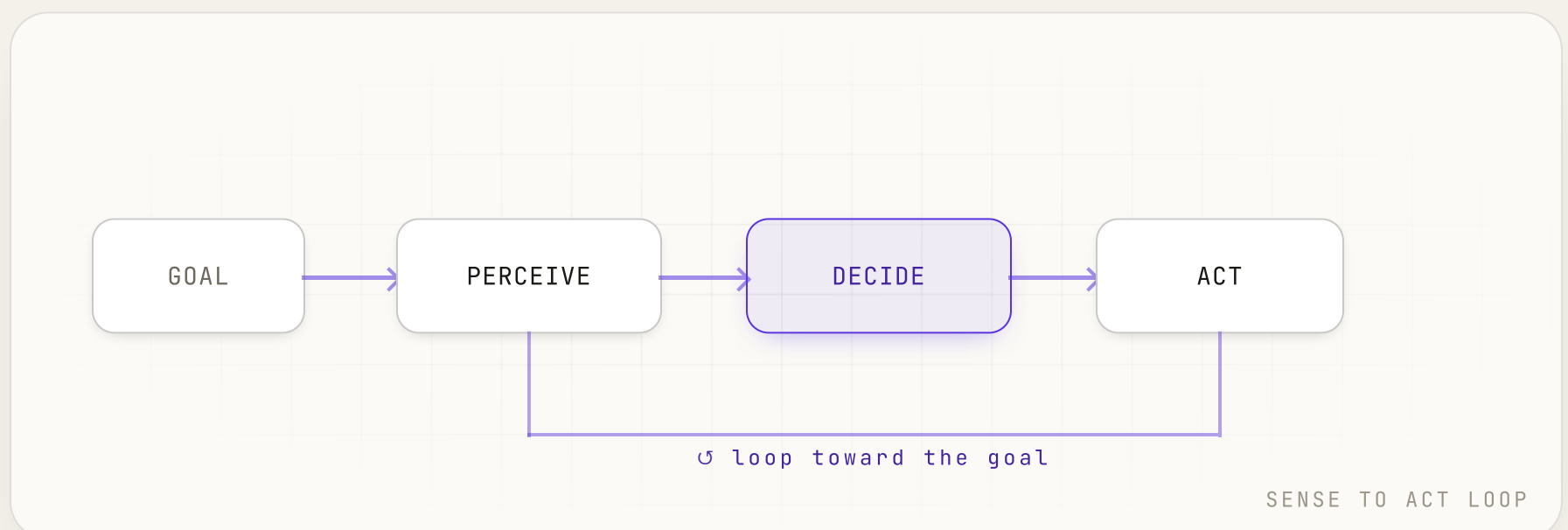
Agent

perceive • decide • act

An agent is an autonomous AI system that **perceives its environment, makes decisions, and takes actions** to achieve specific goals. Unlike passive tools, agents can plan multi-step tasks, adapt to feedback, and operate independently with minimal human intervention.

USED IN

- Tool use
- Automation
- Workflows



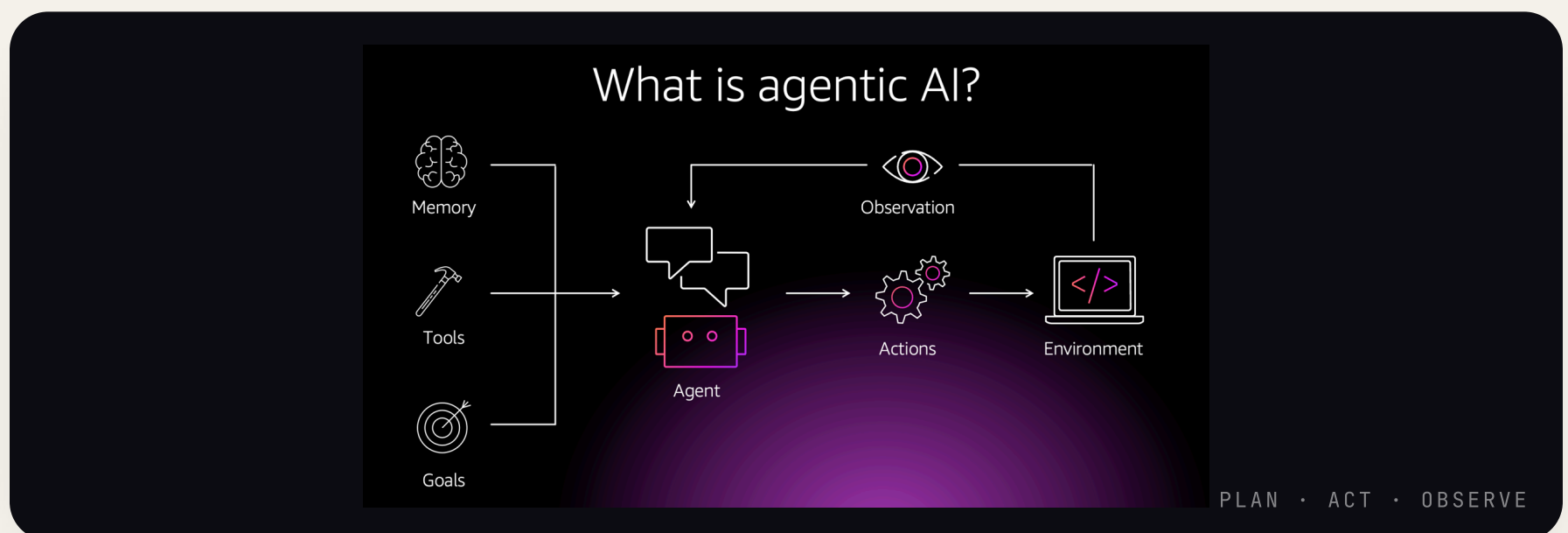
Agentic AI

plan • act • observe • adapt

Agentic AI refers to autonomous AI systems that can **plan, reason, and execute multi-step tasks** independently. Unlike passive language models, agentic AI perceives environments, makes decisions, takes actions, and adapts based on feedback to achieve goals.

USED IN

- Coding agents
- Research
- Ops automation



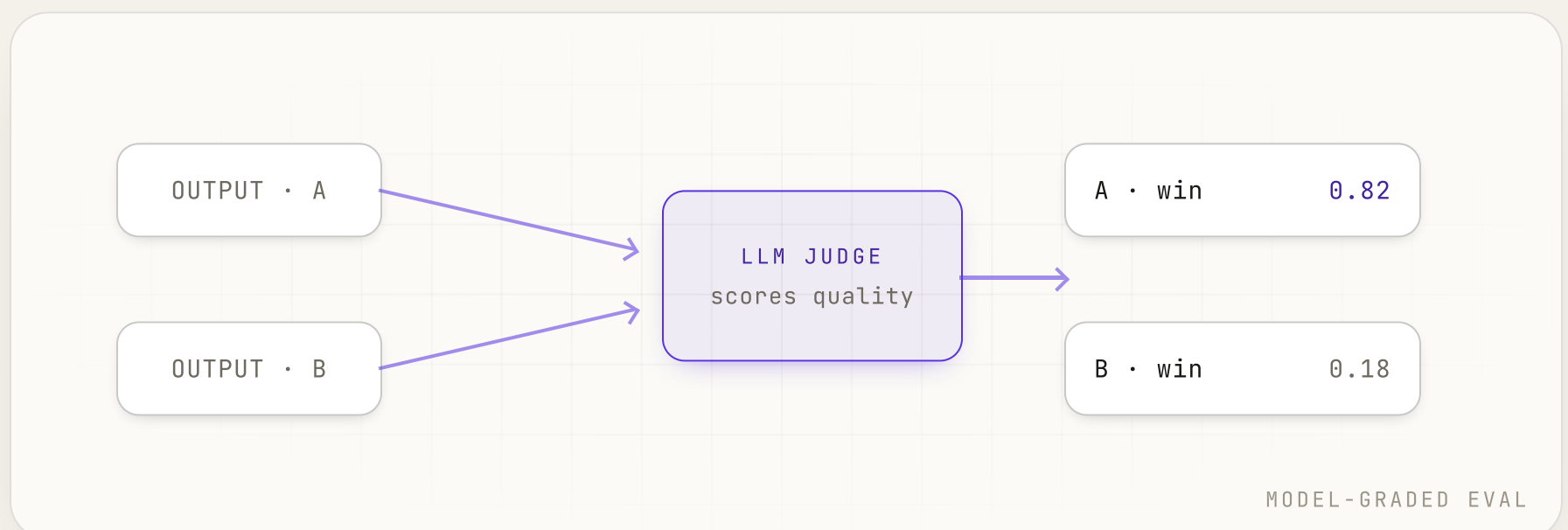
AI as a Judge

model scores model

AI as a Judge uses large language models to **evaluate and score outputs from other systems**, comparing quality, relevance, or correctness. It automates human-like judgment for tasks like ranking responses, grading essays, or assessing model outputs at scale.

USED IN

- Eval pipelines
- Response ranking
- Auto-grading



A/B TESTING • EVALUATION

A/B Test

split • measure • pick

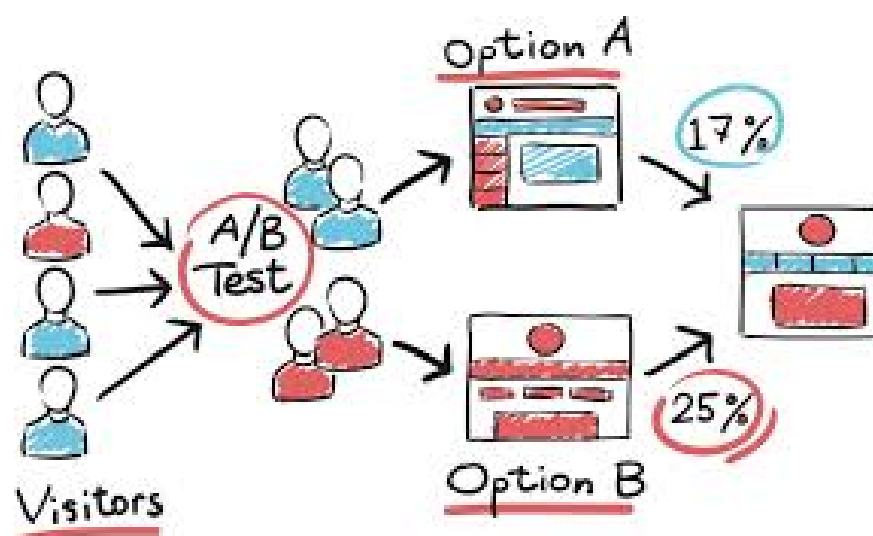
A/B testing compares two versions of something (A and B) by showing them to different groups and **measuring which performs better**. It's a data-driven way to optimize features, designs, or strategies based on real user behavior.

USED IN

• Product

• UX & design

• Growth



CONTROLLED EXPERIMENT

AUC-ROC • EVALUATION

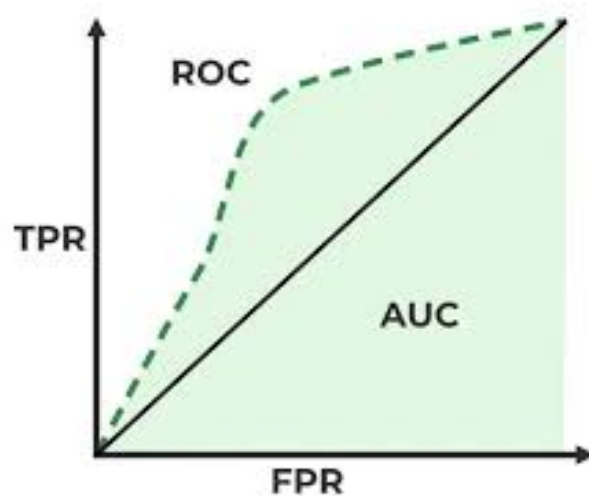
AUC-ROC

TPR vs FPR • area under curve

A single score for how well a classifier **separates classes** across every threshold, plotting true-positive rate against false-positive rate. A value of 1.0 is perfect, 0.5 is a coin flip.

USED IN

- Classifiers
- Model selection
- Imbalanced data



THRESHOLD-FREE METRIC

↓ UP NEXT • TOOL

Tool next.

Beyond the **8 A-words**: the agentic IDE that's rethinking how code gets written.



AGENTIC IDE

Antigravity

agent-first • editor • terminal • browser



Google Antigravity is an agent-first development platform where autonomous AI agents handle complex coding tasks across editor, terminal, and browser simultaneously. Agents plan and execute features end-to-end. Free for individuals with support for Gemini, Claude, and GPT models.

WHAT IT IS

An IDE built around autonomous coding agents.

IN AI STACKS

Agentic coding, planning, tool execution.

WHY DEVS USE IT

Agents handle multi-step tasks, not single edits.

PAIRS WITH

LLMs, terminals, browsers, your repo.

↓ UP NEXT • PEOPLE TO FOLLOW

Spotlight next.

Six **high-signal, low-noise** people behind the A-stack, from the father of AI to today's builders.

A

HIGH-SIGNAL, LOW-NOISE

Alan Turing

1912–1954 • father of AI



Alan Turing laid AI's foundation by formalising computation with the **Turing Machine** and asking "Can machines think?" His 1950 Turing Test became the field's first benchmark for machine intelligence, making him the undisputed father of AI.

KNOWN FOR

The Turing Machine and the Turing Test.

WHY HE MATTERS

Defined what it means for a machine to "think."

HIGH-SIGNAL, LOW-NOISE

Andrew Ng

X • @AndrewYNg



Andrew Ng is a globally recognized AI pioneer who founded [DeepLearning.AI](#), managing general partner at AI Fund, and co-founder of Coursera. With over 8 million learners trained, he has shaped AI education and enterprise adoption worldwide.

FIND HIM

[andrewng.org](#) • @AndrewYNg

BEST FOR

Learning AI from the ground up.

HIGH-SIGNAL, LOW-NOISE

Ashish Vaswani



X • @ashVaswani

Ashish Vaswani is the AI visionary who co-authored the 2017 “Attention Is All You Need” paper, inventing the **Transformer architecture** that powers ChatGPT, Gemini, and modern generative AI. Now leading Essential AI as co-founder and CEO.

FIND HIM

essentialai.com • @ashVaswani

BEST FOR

Transformer-era research and frontier labs.

HIGH-SIGNAL, LOW-NOISE

Aishwarya Srinivasan

@the.datascience.gal



Aishwarya Srinivasan is the **most followed Indian woman in AI** and founder of TheGenAcademy. She democratizes cutting-edge AI and data science education across Instagram, YouTube, and structured bootcamps to 1.1M+ learners globally.

FIND HER

in/aishwarya-srinivasan ·
@the.datascience.gal

BEST FOR

Accessible AI and data science education.

HIGH-SIGNAL, LOW-NOISE

Afshine Amidi

in/afshineamidi



Afshine Amidi co-created the widely shared Stanford CS229 and CS230 cheatsheets, distilling dense machine learning and courses into clean visual references used by learners worldwide. He and his brother also teach Stanford's CME 295 and CME 269 courses, helping students and practitioners build a strong foundation in AI.

FIND HIM

[linkedin.com/in/afshineamidi](https://www.linkedin.com/in/afshineamidi)

BEST FOR

Crisp ML reference notes and cheatsheets.

HIGH-SIGNAL, LOW-NOISE

AI Engineer

YouTube • @aidotengineer



AI Engineer is a YouTube channel and educational platform focused on practical AI development. Through expert interviews, technical deep dives, and panel discussions with researchers, founders, and engineers, it helps practitioners stay current with advances in LLMs, AI agents, RAG, evaluation, and AI product development. The channel serves a global community of over one million AI professionals and bridges cutting-edge research with real-world engineering practice.

FIND IT

ai.engineer • @aidotengineer

BEST FOR

Talks, explainers, and an AI-builder community.

AI RESEARCHER & EDUCATOR

Andrej Karpathy



X • @karpathy

Founding member of OpenAI and former Director of AI at Tesla, leading Autopilot computer vision. Designer of CS231n at Stanford. Now independent, **creating AI education content** including "Neural Networks: Zero to Hero."

WATCH

Neural Networks: Zero to Hero on YouTube

FOLLOW

@karpathy on X, karpathy.ai

The A stack, in one card.

Save it for later. Drop a comment if there's an A-term you'd add.

01 **ANN** neural nets

02 **Activation** non-linearity

03 **Alpha** learning rate

04 **Agent** autonomy

05 **Agentic AI** multi-step

06 **AI as a Judge** llm eval

07 **A/B Testing** experiments

08 **AUC-ROC** classifier score

+ **Antigravity** tool of the letter

@ **7 people to follow** turing • ng • vaswani • karpathy • +3



+
Built by Prasanna

[linkedin.com/in/prasanna-kulkarni-032650180](https://www.linkedin.com/in/prasanna-kulkarni-032650180)

Save it. Share it. See you for B.

AI GLOSSARY • LETTER A • LEVEL 1 UNLOCKED • 19 SLIDES

Up next **B**